Every weighting licenses a plan, no plan satisfies the floors: a quantifier-order separation between compensatory and noncompensatory transition assessment

Amin Abaee[a,*]

[a] *Independent Researcher, Tehran, Iran*

Abstract

Composite indices sort transitions into accepted and rejected under one of two acceptance semantics. Under weight-adaptive (compensatory) acceptance a transition is accepted if every weighting of the index licenses some plan, possibly a different plan for each weighting; under common-plan (noncompensatory) acceptance one plan must keep every criterion above its floor along the whole path. We compare the two as robust finite-horizon certification problems on a common datum with path-wise constraints and worst-case disturbances. At any fixed trajectory the full-cone aggregate loses nothing, yet the semantics separate, because "some plan for every weighting" and "one plan for all weightings" do not commute. For any finite menu of plans governed by required-margin vectors, the acceptance gap is the part of the convex hull of the margins that no single plan dominates, so it vanishes exactly when the menu is convexified. On an explicit rational datum the gap has nonempty interior; its states split, at a closed-form reserve threshold, into those a staged plan can rescue and those no available plan completes safely, with exact weight-ratio thresholds giving the preference range over which each plan is licensed. Blending plans closes the gap; alternating them in time does not. Every positive-elasticity CES index falsely certifies an exact interval of gap states, with critical floors $1 < 5/4 < \sqrt{2} < \varphi < \sqrt{3} < \cdots \to 2$, and a shared shock reduces the gap by an exact cut without removing it. A fishery transition anchored to a regulated biomass limit realises the datum. Where separately binding floors matter, the aggregate alone cannot tell the rescuable from the impossible: per-floor reporting along the path, or a verified common plan, is a detection

*Corresponding author. ORCID: 0000-0002-0019-1842.

Email address: amin_abaee@ut.ac.ir (Amin Abaee)

requirement.

Keywords: multi-criteria decision analysis, compensatory and noncompensatory aggregation, composite indicators, scalarization, robustness, sustainability assessment, viability

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1. Introduction

1.1. Compensability as an acceptance question

Multi-criteria assessment of a sustainability transition faces a choice that the decision-analysis literature has studied since its foundations: whether the criteria may compensate one another. Multi-attribute value theory builds substitution rates between criteria into its trade-off structure (Keeney and Raiffa, 1976); the outranking tradition constrains compensation through concordance and veto (Roy, 1996; Vincke, 1992); and the decision-aiding treatment of sustainable development centres on incommensurable criteria (Munda, 2005; Martinez-Alier et al., 1998). In environmental assessment the same choice appears as the distinction between weak sustainability, in which capital forms are substitutable through prices and an aggregate must not decline (World Bank, 2011; Neumayer, 2013; Boos, 2015), and strong sustainability, in which certain stocks are separately binding (Daly, 1990; Ekins et al., 2003); the indicator theory of the two positions is well developed (Hanley et al., 1999; Martinet, 2011; Cairns and Martinet, 2014; Doyen and Gajardo, 2020; Usubiaga-Liaño, 2025), as is the compensability analysis of multi-criteria methods used for sustainability assessment (Cinelli et al., 2014; Schär et al., 2025).

Nearly all of this work classifies aggregation behaviour at a fixed profile of criterion values. This paper asks a different question, about *transitions*: when an authority must carry out a plan over time, under disturbances, with several criteria required to hold along the entire path, when can a compensatory aggregate certify a transition that a noncompensatory assessment rejects? The question is not about the choice of weights, whose methodology is itself a developed field (Nardo et al., 2008; Becker et al., 2017), and it is not about the known static limitation of weighted sums, which cannot reach nonconvex parts of a Pareto front (Das and Dennis, 1997). It is about the *acceptance semantics* applied to a composite reading of a transition: the rule by which the reading sorts the transition into the accepted or rejected category, the sorting problematic of decision aiding (Roy, 1996).

1.2. Two acceptance semantics

Fix a transition datum: a state carrying typed floors $s_i \geq 0$ (criteria normalised so that zero is the binding level), a finite menu of plans, a disturbance set, and *exact-tube* semantics under which a plan is safe only if every state visited along it, not merely its endpoint, satisfies the constraints. For each nonnegative weighting w in the full cone $W_+ = \mathbb{R}^n_+ \setminus \{0\}$ let $E_w(z)$ be the set of plans admissible at state z under the aggregate floor $w \cdot s \geq 0$, and let $E_{\text{typ}}(z)$ be the set admissible under the typed floors taken separately. Two acceptance criteria differ by quantifier order:

- *common-plan acceptance* (noncompensatory, coordinate-wise): $\exists a \; \forall w : a \in E_w(z)$;
- *weight-adaptive acceptance* (compensatory, scalarized): $\forall w \; \exists a_w : \; a_w \in E_w(z)$.

The first implies the second; the converse fails. Table 1 places the two among the four protocols an index can serve. In the vocabulary of decision making under uncertainty they differ in *when the plan is fixed relative to the weight*: common-plan acceptance commits before the weighting is known, weight-adaptive acceptance lets the plan adapt once it is observed, the here-and-now versus wait-and-see distinction of adjustable robust optimisation (Ben-Tal et al., 2004) with the assessment weight in the role of the uncertain parameter. The separation studied here is Protocol 2 against Protocol 4; it is a statement about a certification protocol applied to a composite index, not an objection to composite indices as such.

A concrete form of the question: a public authority must carry out a restructuring while satisfying two minimum-service constraints. Rapid restructuring exposes one constraint to a temporary drawdown; a gradual plan shifts the burden to the other; a staged plan avoids both but requires a reserve to finance it. An evaluator who aggregates the two constraints into one weighted index may approve the transition for *every* relative price, each price licensing a different plan, while an evaluator who requires each constraint to hold along the whole path approves none of the available plans. The results below show that this divergence is generic in a precise geometric sense, that it is bounded by an exact reserve threshold, that it is closed by blending plans but not by alternating them, and that it survives the replacement of the linear index by any positive-elasticity aggregator.

1.3. Contributions

The paper makes five contributions, stated in the order of the text.

Table 1: Four acceptance protocols for a composite index on a transition datum. The separation studied in this paper is Protocol 2 against Protocol 4; Protocol 3 coincides with Protocol 4 on the full cone (Proposition 1(ii)).

Protocol	Quantifier form	Reading
1. Fixed-weight assessment	$a \in E_w(z)$ at a declared w	one index, one selected plan: ordinary fixed-weight certification
2. Weight-adaptive (per-weight) acceptance	$\forall w\ \exists a_w \in E_w(z)$	every weighting can be paired with some plan, possibly weighting-dependent: the compensatory reading, $\mathcal{V}_{\text{weak}} = \bigcap_w \mathcal{V}_w$
3. Common-plan robustness	$\exists a\ \forall w:\ a \in E_w(z)$	one plan serves every weighting
4. Coordinate-wise (typed) acceptance	$\exists a:\ a \in E_{\text{typ}}(z)$	separately binding floors enforced path-wise: the noncompensatory reading

(i) *Localisation and hierarchy* (Section 3). On the full cone the typed operator is exactly the common-plan intersection, $E_{\text{typ}}(z) = \bigcap_{w \in W_+} E_w(z)$, and at any fixed trajectory the aggregate is lossless (Lemma 2); the chain typed $\subseteq$ scalarized $\subseteq$ physical $\subseteq$ endpoint holds for every weight, and restricting the weight family can only enlarge the compensatory accepted set (Propositions 1–2). The separation is therefore dynamic – a property of the quantifier over plans – and the full cone is its strictest case.

(ii) *A geometric characterisation for any finite menu* (Theorem 1). For plans whose coordinate-wise admissibility is governed by a required-margin vector d_a, weight-adaptive acceptance is $\text{conv}(\mathcal{D}) + \mathbb{R}^n_+$ and common-plan acceptance is $\bigcup_a (d_a + \mathbb{R}^n_+)$; the acceptance gap is the part of the convex hull that no single plan dominates, and convexifying the menu closes it.

(iii) *An explicit rational witness* (Theorem 2). On a four-plan datum the gap is a region with nonempty interior, equal to the *impossibility region* I; the aggregate-versus-floor discrepancy region splits at the reserve level $x = 1$ into I and a *rescue set* R that a reserve-financed staged plan already serves; exact weight-ratio thresholds $\rho_1 = (2 - s_1)/s_2$ and $\rho_2 = s_1/(2 - s_2)$ give the preference range over which each plan is licensed; and the minimal reserve increment that converts impossibility

into rescue is $\kappa^*(z) = (1-x)\,\mathbf{1}[z \notin \mathcal{V}_{\text{typ}}]$ (Proposition 5). The separation persists under hold-prefix extension of the horizon and can be erased by a later stage with coincident terminal sets (Proposition 3).

(iv) *What closes the gap and what does not* (Theorem 3, Proposition 4). Fractional blending of the primitive plans – menu convexification at the level of control flows – makes the typed accepted set coincide exactly with the compensatory region, so the gap is a property of the finite deterministic menu, not of either doctrine; discrete time-sharing closes it only where both dips are separately survivable.

(v) *Robustness of the separation to the aggregator and to the disturbance class* (Theorem 4, Proposition 6). Replacing the linear index by a constant-elasticity-of-substitution (CES) aggregator with elasticity σ changes the gap but does not remove it: the accepted rungs are nested, each positive-elasticity member falsely certifies an exact interval of gap states with critical floors $1 < 5/4 < \sqrt{2} < \varphi < \sqrt{3} < \cdots \to 2$, and only the Leontief member ($\sigma = 0$), which is the typed operator, is uniformly safe. Under a shared environmental shock striking every plan the gap survives, reduced by an exact 1/2-cut, with the licensing thresholds unshifted on the surviving region; the reserve route then fails on an exact band of states.

Every continuum statement is proved (Appendix A–Appendix B; Supplementary Information, SI S2) and every quoted number is an exact rational re-derived by deposited verification code; the core theorems are additionally formalised in Lean 4 (Section 4.7). A resource-transition benchmark (Section 5) realises the datum as a fishery transition with units anchored to a regulated biomass limit.

What is not claimed. No ranking of weak and strong sustainability as doctrines; no proof that any particular set of boundaries ought to be treated noncompensatorily; no empirical result about any resource system; no infinite-horizon, stochastic or partially observed extension. The impossibility region is a negative certificate relative to the specified plan menu, and the separation is exhibited for a declared disturbance class.

1.4. Position in the literature

The backward constructions used here are typed instances of robust-predecessor, capture-basin and reach–avoid constructions (Aubin, 1991; Aubin et al., 2011; Saint-Pierre, 1994; Lygeros et al., 1999; Aubin and

Catté, 2002); the hierarchy of Proposition 1(i) is constraint-set monotonicity of viability kernels (Aubin, 1991; Frankowska, 1989); viability methods are established in natural-resource management (De Lara and Doyen, 2008; Doyen et al., 2012). The blend-collapse pair is the pure-versus-mixed-strategy structure of the assessor–planner game (von Neumann, 1928; Sion, 1958), whose reach–avoid form is the discriminating kernel of Cardaliaguet (1996) (numerically, Cardaliaguet et al., 1999). In composite-indicator practice the critique of weights as value judgements presented as measurement (Hickel, 2020; Fischer et al., 2022) and the construction of indices without weights from Pareto fronts (Gao et al., 2023) both concern the index; the result here is orthogonal – even with weights elicited faithfully, the quantifier order produces the gap. The exactness statement of Lemma 2 is, in the vocabulary of the companion obstruction calculus (Abaee, 2026a), the statement that the typed property is constant on the fibres of the full-cone observation at a fixed trajectory; Theorem 2 shows that this exactness does not survive the quantifier over plans. We do not assert priority over the elementary fact $\exists a\,\forall w \neq \forall w\,\exists a_w$; to our knowledge no explicit exact-tube witness of the dynamic separation in a compensatory/noncompensatory assessment setting has been reported, but whether an equivalent separation appears in adjacent literatures is not established either way.

2. Setting: typed transition data and assessment operators

2.1. Typed transition data and exact-tube semantics

A *typed transition datum* consists of a state space Z whose coordinates carry types (a conserved stock with a unit, a service margin, an institutional label), a finite *menu* $\mathcal{A}$ of plans, a *disturbance class* D, and, for each plan a and disturbance d, a *tube* $\mathrm{Tube}(a,d) \subseteq Z$ – the full set of states visited during one review interval – together with its *endpoint set* $\mathrm{End}(a,d) \subseteq \mathrm{Tube}(a,d)$ and its *successor set* $\mathrm{Succ}(a,d)$, the states reached after any end-of-interval reset. Among the coordinates are n *typed floors* $s = (s_1, \ldots, s_n)$, each normalised so that $s_i \geq 0$ is the binding constraint and each carrying the provenance of its floor (physical, contractual or normative). Typing records which quantities may be added: conservation statements are per type, and the framework does not itself authorise adding biomass, money and biodiversity into one conserved scalar; the scalarization weights studied below are precisely such a bridge, declared as an assessment convention rather than a physical identity.

Two constraint sets complete the datum: a *physical safe set* S^{phys} and a *physical destination set* G^{phys}, which encode constraints that no assessment

doctrine disputes (for instance that a reserve stock cannot be negative). The *typed* sets add the floors,

$$S = S^{\text{phys}} \cap \{s_i \geq 0,\ i = 1, \ldots, n\}, \qquad G = G^{\text{phys}} \cap \{s \geq 0\},$$

and the *scalarized* sets at a weight w add only the aggregate floor,

$$S^w = S^{\text{phys}} \cap \{w \cdot s \geq 0\}, \qquad G^w = G^{\text{phys}} \cap \{w \cdot s \geq 0\}.$$

Throughout, w ranges over the full nonnegative cone $W_+ = \mathbb{R}^n_+ \setminus \{0\}$ unless a subfamily $W \subseteq W_+$ is named. The semantics is *exact-tube*: a plan is admissible only if every visited state, not merely the endpoint, lies in the relevant safe set. The disturbance quantifier is always innermost – admissibility requires safety for every $d \in D$ – and the separation studied here concerns a second quantifier, over weights, placed outside it.

2.2. Assessment operators

For a state z, four operators return the set of plans that an assessment doctrine accepts at z:

$$E_{\text{typ}}(z) = \{a :\ \forall d \in D,\ \text{Tube}(a, d) \subseteq S,\ \text{Succ}(a, d) \subseteq G\}, \tag{1}$$

$$E_w(z) = \{a :\ \forall d \in D,\ \text{Tube}(a, d) \subseteq S^w,\ \text{Succ}(a, d) \subseteq G^w\}, \quad w \in W_+, \tag{2}$$

$$E_{\text{tube,phys}}(z) = \{a :\ \forall d \in D,\ \text{Tube}(a, d) \subseteq S^{\text{phys}},\ \text{Succ}(a, d) \subseteq G^{\text{phys}}\}, \tag{3}$$

$$E_{\text{end}}(z) = \{a :\ \forall d \in D,\ \text{End}(a, d) \subseteq S^{\text{phys}},\ \text{Succ}(a, d) \subseteq G^{\text{phys}}\}. \tag{4}$$

E_{typ} is the *noncompensatory typed* operator (each floor separately binding along the path); E_w the *scalarized* operator at weight w (the dashboard reading $w \cdot s$ must stay nonnegative along the path); $E_{\text{tube,phys}}$ the *physical* operator (no floors at all); and E_{end} the *endpoint* operator, which represents any assessment architecture that evaluates constraints only at sampled or audited endpoints. The two axes along which they differ – what is constrained and where it is evaluated – are set out in Table 2; the typed-endpoint operator $E_{\text{end,typ}}$, which applies the floors at endpoints only, is the remaining cell and is used once, in Section 6.3.

2.3. Accepted-state sets and the central inclusion

For any operator E write $\mathcal{V}[E] = \{z :\ E(z) \neq \varnothing\}$ for its *accepted-state set*, the states at which the doctrine admits at least one plan. Thus

Table 2: Evaluation semantics: constraint structure (rows) by evaluation locus (columns).

	Evaluated on the full tube	Evaluated at endpoints only
Typed floors	E_{typ} – typed path safety	$E_{\text{end,typ}}$ – typed snapshots
Scalarized floor	E_w – scalarized path safety	(not isolated here)
Physical constraints only	$E_{\text{tube,phys}}$ – physical viability	E_{end} – audited physical snapshots

$\mathcal{V}_{\text{typ}} = \mathcal{V}[E_{\text{typ}}]$, $\mathcal{V}_w = \mathcal{V}[E_w]$, $\mathcal{V}_{\text{phys}} = \mathcal{V}[E_{\text{tube,phys}}]$, $\mathcal{V}_{\text{end}} = \mathcal{V}[E_{\text{end}}]$, and the *compensatory accepted set* is the per-weight intersection

$$\mathcal{V}_{\text{weak}} = \bigcap_{w \in W_+} \mathcal{V}_w = \{z : \ \forall w \ \exists a_w \in E_w(z)\}. \tag{5}$$

The object of the paper is the inclusion

$$\underbrace{\{z : \ \textstyle\bigcap_w E_w(z) \neq \varnothing\}}_{\text{common plan}} \ \subseteq \ \underbrace{\textstyle\bigcap_w \{z : \ E_w(z) \neq \varnothing\}}_{\text{per weight } = \mathcal{V}_{\text{weak}}}, \tag{6}$$

whose left-hand side will turn out to be exactly $\mathcal{V}_{\text{typ}}$ (Proposition 1(ii)), and the set $\text{FP}_{\text{agg}} = \mathcal{V}_{\text{weak}} \setminus \mathcal{V}_{\text{typ}}$ on which it is strict – the *acceptance gap*. Table 3 collects the symbols.

3. General results: localisation, hierarchy and the finite-menu geometry

3.1. Two lemmas on quantifier order

The first lemma is the abstract form of the separation; the second shows that it cannot be a static phenomenon.

Lemma 1 (quantifier inclusion)**.** *Let $\{E_\lambda(z)\}_{\lambda \in \Lambda}$ be any family of plan sets. Then*

$$\left\{z : \ \textstyle\bigcap_\lambda E_\lambda(z) \neq \varnothing\right\} \ \subseteq \ \textstyle\bigcap_\lambda \left\{z : \ E_\lambda(z) \neq \varnothing\right\},$$

with equality exactly when, for every z in the right-hand set, the family $\{E_\lambda(z)\}_\lambda$ admits a common selector. No topology on the plan space is needed; for a finite menu the check is a finite enumeration.

Lemma 2 (the full cone is lossless at a fixed trajectory)**.** *For $v \in \mathbb{R}^n$: $v \geq 0$ componentwise if and only if $w \cdot v \geq 0$ for every $w \in W_+$.*

Table 3: Notation.

Symbol	Meaning
z; $a \in \mathcal{A}$; $d \in D$	state; plan (action) from the finite menu; disturbance from the declared class
$s = (s_1, \ldots, s_n)$; x	typed floors, normalised so that $s_i \geq 0$ binds; reserve (bridge) stock on the witness
$W_+ = \mathbb{R}^n_+ \setminus \{0\}$; $W \subseteq W_+$; $r = w_2/w_1$	full weight cone; a weight subfamily; weight ratio on the two-floor witness
S, G; S^w, G^w; $S^{\text{phys}}, G^{\text{phys}}$	typed, scalarized and physical safe/destination sets
$\text{Tube}, \text{End}, \text{Succ}$	visited set, endpoint set and successor set of a plan under a disturbance
E_{typ}, E_w, $E_{\text{tube,phys}}$, E_{end}	assessment operators (1)–(4); $\mathcal{V}[E] = \{z : E(z) \neq \varnothing\}$
$\mathcal{V}_{\text{typ}}$, $\mathcal{V}_w$, $\mathcal{V}_{\text{weak}}$, $\mathcal{V}_{\text{phys}}$, $\mathcal{V}_{\text{end}}$	accepted-state sets; $\mathcal{V}_{\text{weak}} = \bigcap_{w \in W_+} \mathcal{V}_w$
$\text{FP}_{\text{agg}} = \mathcal{V}_{\text{weak}} \setminus \mathcal{V}_{\text{typ}}$	the acceptance gap
d_a; $\mathcal{D} = \{d_a\}$	required-margin vector of plan a (Theorem 1)
$\mathcal{Q}$, R, I; ρ_1, ρ_2	discrepancy region, rescue set, impossibility region; licensing thresholds (Theorem 2)
$e = (\frac{1}{4}, \frac{1}{4})$; $c = 1$; κ, κ^*	destination gain and rescue cost of the witness; reserve increment and rescue threshold
θ; $\sigma = 1/(1-\theta)$; $\lambda_i = 1 + s_i$	CES exponent, elasticity of substitution, floor-referenced index (Theorem 4)

Proofs are in Appendix A. Lemma 2 says that an assessor who checks the aggregate floor $w \cdot s \geq 0$ at *every* nonnegative weighting, along a *fixed* trajectory, detects exactly the same violations as an assessor who checks each floor separately. Whatever separates the two doctrines must therefore enter through the quantifier over plans in (6), not through any blindness of scalarization at a profile.

3.2. Hierarchy, localisation and weight-family monotonicity

Proposition 1 (hierarchy and localisation)**.** (i) *For every $w \in W_+$ and every z,*

$$E_{\text{typ}}(z) \;\subseteq\; E_w(z) \;\subseteq\; E_{\text{tube,phys}}(z) \;\subseteq\; E_{\text{end}}(z).$$

(ii) *For every z, $E_{\text{typ}}(z) = \bigcap_{w \in W_+} E_w(z)$; hence $\mathcal{V}_{\text{typ}} = \{z : \bigcap_w E_w(z) \neq \varnothing\}$, and the central inclusion (6) reads $\mathcal{V}_{\text{typ}} \subseteq \mathcal{V}_{\text{weak}}$.*

Part (i) is constraint-set monotonicity under nested safe sets, the standard monotonicity of viability kernels (Aubin, 1991; Frankowska, 1989). Part (ii) is the localisation that gives the paper its subject: the noncompensatory doctrine *is* common-plan robustness over the full cone (Protocols 3 and 4 of Table 1 coincide), so the only difference between the doctrines is the order of the two quantifiers.

Proposition 2 (weight-family monotonicity)**.** *For a weight family $W \subseteq W_+$ let $\mathcal{V}_W = \bigcap_{w \in W} \mathcal{V}_w$. Then $\mathcal{V}_{\text{typ}} \subseteq \mathcal{V}_W \subseteq \mathcal{V}_{\text{phys}}$, and $W_1 \subseteq W_2$ implies $\mathcal{V}_{W_2} \subseteq \mathcal{V}_{W_1}$.*

Restricting the weight family – eliciting a narrower set of admissible weightings – can only *enlarge* the compensatory accepted set. The full cone is thus the strictest compensatory reading, and a separation exhibited at the full cone persists for every restricted family; the elicitation of weights, however careful, cannot remove it.

3.3. The finite-menu geometric characterisation

The mechanism behind the gap is geometric. Call a plan a *margin-class* plan if its coordinate-wise admissibility is governed by a *required-margin vector* $d_a \in \mathbb{R}^n_+$: plan a is admissible at floor margins $s \geq 0$ exactly when $d_a \leq s$ componentwise, and it satisfies the destination constraint. A plan whose worst-case tube dips each floor i by at most $d_{a,i}$ and returns is of this class; plans whose admissibility depends on another resource (reserve-financed plans) or that fail the destination are handled separately, as on the witness of Section 4.

Theorem 1 (finite-menu geometric characterisation)**.** *Let $\mathcal{A}$ be a finite menu of margin-class plans with required margins $\mathcal{D} = \{d_a : a \in \mathcal{A}\}$. Then*

(i) *coordinate-wise acceptance is* $\{s : \exists a,\ d_a \le s\} = \bigcup_{a\in\mathcal{A}}(d_a + \mathbb{R}^n_+)$*;*

(ii) *scalarized (weight-adaptive) acceptance is* $\{s : \forall w \in W_+\ \exists a,\ w \cdot d_a \le w \cdot s\} = \mathrm{conv}(\mathcal{D}) + \mathbb{R}^n_+$*.*

Hence the acceptance gap is $\left[\mathrm{conv}(\mathcal{D}) + \mathbb{R}^n_+\right] \setminus \bigcup_a (d_a + \mathbb{R}^n_+)$ *– the part of the convex hull of the required margins that no single plan dominates – and replacing the menu by its convex hull, i.e. admitting every convexified plan, collapses the gap to the empty set.*

The proof (Appendix A) is a separating-hyperplane argument; its content is displayed in Figure 1. Weight-adaptive acceptance of s means that s is not strictly separable from the up-closed convex hull of the required margins – for every weighting some plan's margin is cheaper than s in that weighting – while coordinate-wise acceptance means that s dominates some margin outright. The two coincide exactly when the up-closure of $\mathcal{D}$ is already convex, in particular whenever one plan dominates all others or the menu is convexified. Two consequences organise the rest of the paper: the gap is a property of the *finiteness and nonconvexity of the menu*, not of either doctrine; and it is closed by convexifying the menu (Theorem 3), whereas instruments that act on the weights alone cannot close it (Proposition 2). The nonconvexity at work is menu geometry; it is distinct from the Pareto-front nonconvexity under which weighted sums fail in multi-objective optimisation (Das and Dennis, 1997), which acts through frontier geometry under a single optimisation and has no quantifier over plans.

4. An explicit witness: the separation on a rational datum

4.1. The witness datum

Two architectures – extraction ($q = 0$) and regeneration ($q = 1$) – share one review interval $[0, 1]$. The phase state is (q, x, s_1, s_2): a physical reserve stock x (a bridge stock, for instance an adjustment fund) and two typed floors s_1 and s_2 normalised at zero (on the benchmark of Section 5: a biomass margin above a limit reference point and a fleet income margin). The transition-safe set is $S_0 = \{x \ge 0,\ s_1 \ge 0,\ s_2 \ge 0\}$ and the destination set is $G = \{(1, x, s) : x \ge 0,\ s \ge 0\}$, with destination maintainability witnessed by a hold policy. The destination reset adds the gain $e = (\frac{1}{4}, \frac{1}{4})$ to both floors; e is strictly positive, so successors are interior to G and no

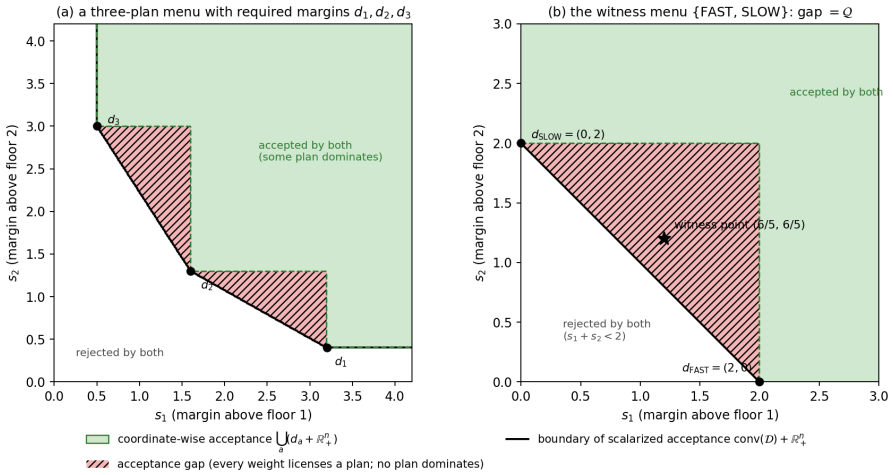


Figure 1: The finite-menu geometric characterisation (Theorem 1) in the plane of two floor margins. Green: coordinate-wise acceptance, the union of the orthants $d_a + \mathbb{R}^2_+$ (some plan dominates). Black boundary: scalarized acceptance, $\mathrm{conv}(\mathcal{D}) + \mathbb{R}^2_+$. Hatched: the acceptance gap, where every weighting licenses some plan but no single plan keeps every floor. (a) A three-plan menu. (b) The margin-class plans of the witness datum, $d_{\mathrm{FAST}} = (2, 0)$ and $d_{\mathrm{SLOW}} = (0, 2)$: the gap is the discrepancy triangle $\mathcal{Q} = \{s_1 < 2,\ s_2 < 2,\ s_1 + s_2 \geq 2\}$ of Theorem 2, with the witness point $(6/5, 6/5)$ in its interior; the reserve-financed plan, which is outside the margin class, serves the triangle only when the reserve covers its cost.

inequality below binds on its magnitude. The rescue cost is $c = 1$. Initial states are $X_0 = \{(0, x, s) : x \geq 0,\ s \geq 0\}$.

Four plans are available (Table 4), written as piecewise-linear maps on the breakpoints $\{0, \frac{1}{2}, 1\}$ and monotone on each piece, so that every tube is the exact visited set. The disturbance class is *action-indexed*: the adverse disturbance triggers the worst-case dip, of fixed depth 2, in the active plan's characteristic coordinate (s_1 for FAST, s_2 for SLOW); on the benchmark the depth 2 is a quota pulse of 3/2 plus an environmental shock of 1/2. The worst-case tube of FAST is therefore $[s_1 - 2, s_1]$ in s_1 with s_2 and x constant, that of SLOW is symmetric, and that of STAGED is $[x - 1, x]$ in x with both floors nondecreasing. A shared-shock class, in which the environmental component strikes every plan, is treated in Section 4.6.

Table 4: The plan menu of the witness datum from an initial state $(0, x, s)$, $x \geq 0$, $s \geq 0$. Trajectories are worst-case; $e = (\frac{1}{4}, \frac{1}{4})$.

Plan	Within-interval trajectory (worst case)	Successor
NO-SWITCH	state constant	$\{(0, x, s)\}$ – misses G
FAST	$s_1(t) = s_1 - 4t$ on $[0, \frac{1}{2}]$, $s_1 - 4(1-t)$ on $[\frac{1}{2}, 1]$; s_2, x constant	$\{(1, x, s + e)\}$
SLOW	$s_2(t) = s_2 - 4t$ on $[0, \frac{1}{2}]$, $s_2 - 4(1-t)$ on $[\frac{1}{2}, 1]$; s_1, x constant	$\{(1, x, s + e)\}$
STAGED	$x(t) = x - t$ on $[0, 1]$; floors grow linearly to $s + e$	$\{(1, x - 1, s + e)\}$

The datum is deliberately spare. Its margin-class plans are FAST and SLOW, with $\mathcal{D} = \{(2, 0), (0, 2)\}$, so Theorem 1 predicts the gap $\{s_1 + s_2 \geq 2\} \setminus (\{s_1 \geq 2\} \cup \{s_2 \geq 2\})$ among them; STAGED lies outside the margin class, because its admissibility is governed by the reserve rather than by the floors, and it is what splits the predicted gap into a rescuable and an impossible part.

4.2. The witnessed separation

Theorem 2 (witnessed separation)**.** *On the witness datum, over X_0:*

(1) $\mathcal{V}_{\text{typ}} = \{x \geq 1\} \cup \{s_1 \geq 2\} \cup \{s_2 \geq 2\}$.

(2) $\mathcal{V}_{\text{weak}} = \bigcap_{w \in W_+} \mathcal{V}_w = \{x \geq 1\} \cup \{s_1 + s_2 \geq 2\}$.

(3) $\mathcal{V}_{\text{phys}} = \mathcal{V}_{\text{end}} = X_0$.

Define the discrepancy region $\mathcal{Q} = \{s_1 < 2,\ s_2 < 2,\ s_1 + s_2 \geq 2\}$ *(x unrestricted), the* rescue set $R = \mathcal{Q} \cap \{x \geq 1\}$ *and the* impossibility region $I = \mathcal{Q} \cap \{x < 1\}$. *Then:*

(4) $\mathrm{FP}_{\mathrm{agg}} = \mathcal{V}_{\mathrm{weak}} \setminus \mathcal{V}_{\mathrm{typ}} = I$. *The acceptance gap is exactly the impossibility region; it is nonempty, with relative interior* $\{0 < x < 1,\ 0 < s_1 < 2,\ 0 < s_2 < 2,\ s_1 + s_2 > 2\}$ *in the* $q = 0$ *slice. The rescue set* R *lies in* $\mathcal{V}_{\mathrm{typ}}$*, not in the gap.*

(5) *Both hierarchy inclusions are strict: every point of* I *lies in* $\mathcal{V}_{\mathrm{weak}} \setminus \mathcal{V}_{\mathrm{typ}}$*, and* $(x, s_1, s_2) = (\frac{1}{2}, \frac{1}{10}, \frac{1}{10})$ *lies in* $\mathcal{V}_{\mathrm{phys}} \setminus \mathcal{V}_{\mathrm{weak}}$.

(6) (Per-weight plan disagreement.) *On the relative interior of* $\mathcal{Q}$ *(*$0 < s_2 < 2$*), with* $r = w_2/w_1$*,* FAST *is licensed exactly for* $r \geq \rho_1$ *and* SLOW *exactly for* $r \leq \rho_2$*, where*

$$\rho_1 = \frac{2 - s_1}{s_2}, \qquad \rho_2 = \frac{s_1}{2 - s_2}, \qquad \rho_2 \geq \rho_1 \iff s_1 + s_2 \geq 2.$$

Assessors with $r > \rho_2$ *license* FAST *only, those with* $r < \rho_1$ *license* SLOW *only, intermediate assessors license both. On* I *no single plan serves every weight:* $\bigcap_w E_w(z) = E_{\mathrm{typ}}(z) = \varnothing$. *On* R*,* STAGED *is typed-admissible and serves every weight.*

(7) (The rescue split.) R *is typed-transformable, witnessed by* STAGED*: bridging at cost* $c = 1$ *keeps both floors intact and lands in* G. I *is aggregate-feasible for every cone weight yet admits no typed-admissible plan, with four exhibited violations:* FAST *breaches the* s_1 *floor under its adverse disturbance,* SLOW *the* s_2 *floor,* STAGED *drives* x *negative, and* NO-SWITCH *misses* G.

The proof is in Appendix B; the formulas for ρ_1, ρ_2 require $0 < s_2 < 2$, and on the boundary faces the classification follows from the closed-set identities (1)–(2). Table 5 records the two verdicts region by region, and Figures 2 and 3 display them.

The witness point. At $(x, s_1, s_2) = (\frac{1}{2}, \frac{6}{5}, \frac{6}{5})$ the typed verdict is rejection: STAGED's reserve tube reaches $\frac{1}{2} - 1 < 0$, FAST's worst-case s_1 dips to $\frac{6}{5} - 2 = -\frac{4}{5}$, SLOW's s_2 likewise, and NO-SWITCH misses G. Yet $s_1 + s_2 = \frac{12}{5} \geq 2$, so the state is compensatorily accepted, with thresholds $\rho_1 = \frac{2}{3}$ and $\rho_2 = \frac{3}{2}$: $r = 2$ licenses FAST only, $r = \frac{1}{2}$ licenses SLOW only, $r = 1$ licenses both. Every weighting licenses a plan; no plan is licensed by every weighting. With the same floors and $x = 1$ the state lies in R: STAGED's tube is $[0, 1]$, the floors grow, and the successor $(1, 0, s + e) \in G$.

Reading. The compensatory assessment's binding condition is a total budget, $s_1+s_2 \geq 2$; the noncompensatory assessment requires either one floor to survive its own worst-case dip ($s_i \geq 2$) or the reserve to cover the rescue cost ($x \geq 1$). Between the two conditions the compensatory doctrine certifies – per weight, with weight-dependent plans – transitions the noncompensatory doctrine rejects, and the aggregate reading cannot tell I from R: the two differ only in x, which the index does not see. Part (7) is a *negative certificate* in the sense of a rejection with an exhibited violated constraint for each plan, exhausting the menu; it is a complete verdict relative to the menu, not an inconclusive one.

Table 5: Region dictionary on the witness datum (Theorem 2).

Region	Per-weight verdict	Common-plan verdict	Reason
$x \geq 1$, all s	accept	accept	STAGED serves every weight
$s_1 \geq 2$ or $s_2 \geq 2$	accept	accept	FAST/SLOW survives its own dip
R: $x \geq 1$, $s_1 < 2$, $s_2 < 2$, $s_1 + s_2 \geq 2$	accept	accept	the reserve finances STAGED
I: $x < 1$, $s_1 < 2$, $s_2 < 2$, $s_1 + s_2 \geq 2$	accept (weight-dependent plans)	reject	dips breach the floors; the reserve cannot finance phasing
$x < 1$, $s_1 + s_2 < 2$	reject	reject	even the index fails

4.3. Persistence under horizon extension, and its limit

Is the one-interval separation an artefact of the one-interval framing? Two results delimit the answer.

Proposition 3 (persistence and erasure)**.** (i) *Extend the witness datum to $m \geq 2$ review intervals by prepending hold intervals, with sole prefix action* HOLD *(tube $\{z\}$, successor $\{z\}$, safe set S_0) available at every prefix state, unchanged terminal and successor architecture, and no additional reset or disturbance branches. Then for every stage j, $\mathcal{V}_j^{\mathrm{typ}} \subseteq \bigcap_w \mathcal{V}_j^w \subseteq \mathcal{V}_j^{\mathrm{phys}}$, the stage-0 accepted regions are the witness regions pulled back through the holds, and both strictness witnesses of Theorem 2(5) persist.* (ii) *Persistence is architecture-conditional: there is a two-interval datum whose single-interval truncation exhibits an acceptance gap and whose two-stage extension erases*

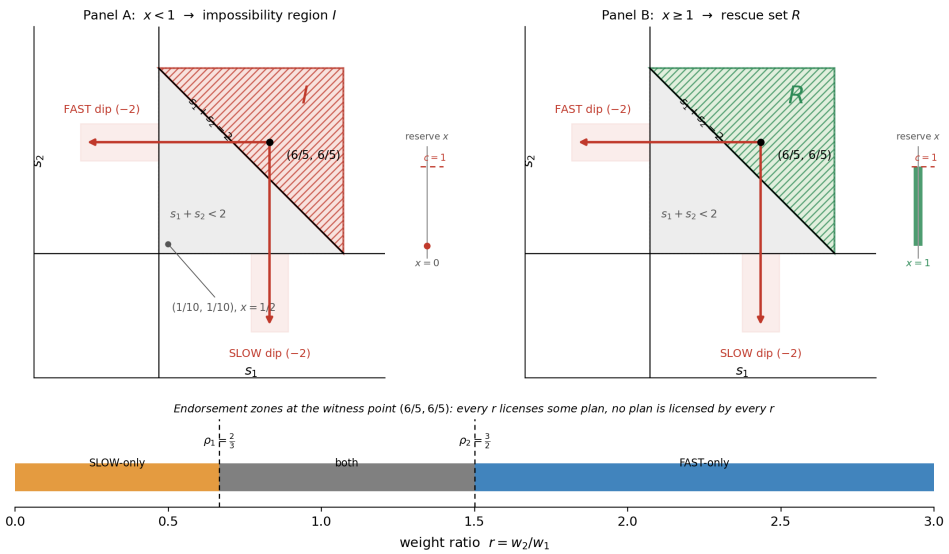


Figure 2: The discrepancy triangle $\mathcal{Q} = \{s_1 < 2,\ s_2 < 2,\ s_1 + s_2 \geq 2\}$ in the floor plane. Panel A ($x < 1$): $\mathcal{Q}$ is the impossibility region $I = \mathrm{FP}_{\mathrm{agg}}$; the arrows show the worst-case dips of FAST ($s_1 \to s_1 - 2$) and SLOW ($s_2 \to s_2 - 2$) crossing the floors from the witness point $(6/5, 6/5)$, and the point $(1/10, 1/10)$ is the second strictness witness. Panel B ($x \geq 1$): the same triangle is the rescue set R, served by STAGED. Below the diagonal both assessments reject when $x < 1$ and accept when $x \geq 1$. The reserve bars show the bridge stock against the rescue cost $c = 1$ at representative slices ($x = 0$ and $x = 1$; the witness point sits at $x = \frac{1}{2}$). The bottom strip gives the licensing zones at the witness point: SLOW only for $r < \rho_1 = \frac{2}{3}$, both for $\frac{2}{3} \leq r \leq \frac{3}{2}$, FAST only for $r > \rho_2 = \frac{3}{2}$.

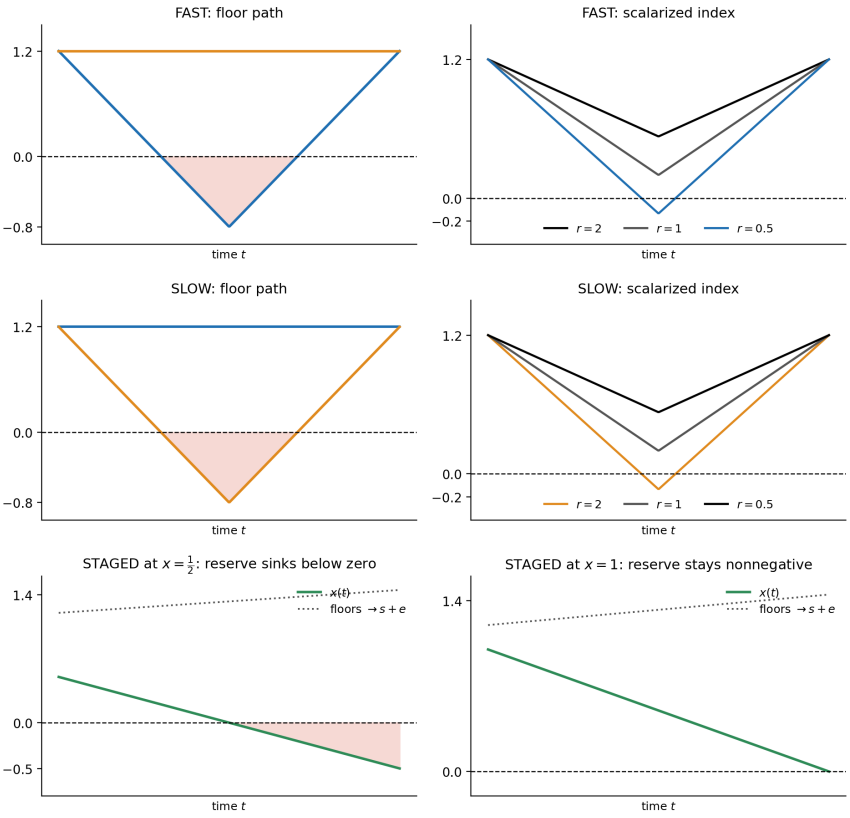


Figure 3: Path view at the witness point under the worst-case disturbance. Rows: FAST, SLOW, STAGED. Left: the floor paths (solid) against the zero floor (dashed). Right: the scalarized index $s_1 + rs_2$ for $r \in \{2, 1, \frac{1}{2}\}$. Under FAST the index stays nonnegative for $r = 2$ and $r = 1$ while s_1 breaches its floor; under SLOW the mirror holds. The only plan that protects both floors, STAGED, consumes the reserve linearly: infeasible at $x = \frac{1}{2}$, feasible at $x = 1$ – the difference between I and R.

it, because the second stage has coincident typed and scalarized terminal sets and a repair action admissible from every state.

The erasure datum of (ii) (SI S2.2) has two floors, no reserve, a first stage with actions that move one unit from one floor to the other, and a second stage with a universal repair action; at $(\frac{2}{5}, \frac{2}{5})$ the truncation has the gap (the two first-stage actions are licensed for $r \geq \frac{3}{7}$ and $r \leq \frac{7}{3}$ respectively, and neither alone is typed-safe), while the extension accepts the state under both doctrines. The witness datum itself is not subject to this erasure: its gap is tube-driven – on I every plan violates the stage-local path constraint S_0 – and a later stage cannot repair a violated tube. Together the two parts state exactly what is claimed: hold-prefix extension preserves the separation; arbitrary multi-stage extension does not; and the erasure mechanisms (a common typed-safe continuation for every aggregate-feasible state, coincident terminal sets on the reachable set, or actions coupled across stages) are realisable.

4.4. What closes the gap and what does not

Theorem 1 predicts that convexifying the menu closes the gap. On the witness datum the convexification instrument is the *blend family*: for $\delta \in (0,1)$, $\textsc{blend}_\delta$ is the plan obtained by fractional allocation of the primitive control flows, $u_\delta = \delta\, u_{\text{FAST}} + (1-\delta)\, u_{\text{SLOW}}$; by linearity of the flow map in the control, its worst-case tube is the pointwise convex combination of the two primitive worst-case tubes and its successor the convex combination of the two successors. This is menu convexification at the level of control inputs – an explicit model assumption, which must be demonstrated from the model equations or carried as a management assumption (fractional allocation of effort, quota or season, with monitoring and enforcement) before the theorem is applied to a real menu.

Theorem 3 (blend collapse)**.** *On the witness datum, over X_0:*

(i) *$\textsc{blend}_\delta$ is typed-admissible at z exactly when $\delta \in \left[1 - \frac{s_2}{2},\ \frac{s_1}{2}\right] \cap [0,1]$, an interval nonempty exactly when $s_1 + s_2 \geq 2$ (a singleton at equality), independent of $x \geq 0$ and of the weight;*

(ii) *the typed accepted set of the menu augmented by the blend family is exactly the compensatory region, $\mathcal{V}_{\text{typ}}^{\text{blend}} = \{x \geq 1\} \cup \{s_1 + s_2 \geq 2\} = \mathcal{V}_{\text{weak}}$: the gap vanishes, and the blend family reaches nothing beyond $\mathcal{V}_{\text{weak}}$;*

(iii) *the admissible window depends on the state through* (s_1, s_2) *alone, so a single* δ *serves every weight: on* I *the blend is the common plan that no member of the finite menu supplies.*

Proposition 4 (discrete time-sharing does not close the gap)**.** *Let a sequential policy alternate between* FAST *and* SLOW *over the interval, each held at full strength on its sub-intervals, so that the visited set is the union of the two primitive worst-case tubes. Such a policy is typed-admissible if and only if* $s_1 \geq 2$ *and* $s_2 \geq 2$*; on* I *none is, and the gap survives.*

The pair delimits "mixing" precisely. Continuous convexification – fractional allocation of flows – closes the gap exactly at the compensatory region $s_1 + s_2 \geq 2$; discrete alternation at plan granularity closes it only where both dips are separately survivable. The structure is that of pure versus mixed strategies in the assessor–planner game (von Neumann, 1928; Sion, 1958): with the planner choosing a plan, the assessor a weight, and the payoff the worst-case aggregate, common-plan acceptance is the pure-strategy value, per-weight acceptance the mixed value, and the gap the pure-strategy duality gap that the convex extension closes; its reach–avoid form is the discriminating kernel of Cardaliaguet (1996). Proposition 4 is scoped to worst-case alternation at finite rate; infinitesimally fast switching recovers the convexified trajectory in the sense of relaxed controls (Filippov, 1988; Warga, 1972) and is subsumed by Theorem 3, not distinct from it.

The remaining instrument acts on the reserve. Define the augmentation map $\mathsf{Aug}_\kappa : (x, s, \mathcal{A}) \mapsto (x + \kappa,\ s,\ \mathcal{A} \cup \{\text{STAGED}_\kappa\})$, where STAGED_κ is STAGED run from the augmented reserve, and the *rescue threshold* $\kappa^*(z) = \inf\{\kappa :\ \exists a \in \mathcal{A}_\kappa,\ a \in E_{\text{typ}}(z)\}$.

Proposition 5 (rescue threshold)**.** *On the witness datum, for every* $z \in X_0$*,*

$$\kappa^*(z) = (1 - x)_+ \cdot \mathbf{1}[\, x < 1,\ s_1 < 2,\ s_2 < 2\,] = (1 - x) \cdot \mathbf{1}[\, z \notin \mathcal{V}_{\text{typ}}\,].$$

On I the threshold is $1 - x > 0$; on R and on all of $\mathcal{V}_{\text{typ}}$ it is 0. Because STAGED_κ is admissible exactly when $x + \kappa \geq 1$, independently of the floors, the reserve route admits every state of the failure set $\{x < 1, s_1 < 2, s_2 < 2\}$ at cost $1 - x$ – including states such as $(\frac{1}{2}, \frac{1}{10}, \frac{1}{10})$ that even the aggregate rejects; I is the part of the failure set on which the aggregate still certifies. Along this route the distance to the feasible slice $\{x \geq 1\}$ equals the violation $(1 - x)_+$ itself, so the error-bound modulus (Fabian et al., 2010) of the relation is identically 1: the route is perfectly conditioned. Table 6 summarises how the instruments act on the gap.

Table 6: Instruments acting on the acceptance gap.

Instrument	Operation	Effect on the gap
Reweighting	change w	none: the gap is a statement over all w
Restricting the weight family	replace W_+ by $W \subset W_+$	enlarges per-weight acceptance; leaves the common-plan problem (Proposition 2)
Reserve augmentation	add STAGED_κ, $x + \kappa \geq 1$	closes it where only the reserve binds, at cost $\kappa^* = 1 - x$ (Proposition 5)
Menu convexification	replace $\mathcal{D}$ by $\text{conv}(\mathcal{D})$ (blend family)	closes it exactly on the scalarized region (Theorems 1, 3)
Discrete time-sharing	union of the primitive tubes	closes it only where both dips are separately survivable (Proposition 4)

4.5. Beyond linear aggregation: the substitutability ladder

The results so far concern the linear index. A natural objection is that the gap is an artefact of perfect substitutability and would close under an aggregator with limited substitution. We evaluate Protocol 2 under the CES family. Work in floor-referenced indices $\lambda_i = 1 + s_i$ ($\lambda_i = 1$ at the floor, $\lambda_i \leq 0$ a full floor-unit below it, the currency of geometric-mean abundance indices) and, for weights normalised to sum to one, the power means

$$M_\theta(\lambda; w) = \Big(\sum_i w_i \lambda_i^\theta \Big)^{1/\theta}, \qquad \sigma = \frac{1}{1-\theta},$$

with the usual limits: $\theta = 1$ is the linear aggregate, $\theta = 0$ the geometric mean (the functional form of the Living Planet Index), $\theta = -1$ the harmonic mean, $\theta \to -\infty$ the Leontief minimum. A state is *accepted at rung* θ when, for every weight, some plan of the menu keeps $M_\theta \geq 1$ along its worst-case tube; the physical and destination constraints are aggregator-independent (STAGED needs $x \geq 1$ and then serves every weight at every rung). Two conventions fix the family. With normalised weights $\sum_i w_i \lambda_i \geq 1 \iff w \cdot s \geq 0$, so the $\theta = 1$ member reproduces the compensatory operator of Section 2 exactly, with the same licensing thresholds (the *handshake*). For every $\theta < 1$ a tube value $\lambda_i \leq 0$ rejects the plan (the *collapse convention*: the geometric mean is 0 there and the $\theta < 0$ members are undefined there); only the linear member compensates across a collapsed coordinate.

On the gap-region diagonal $z = (x < 1, s, s)$, $s \in (1, 2)$, the per-weight cover condition reduces, by the symmetry of the two principal plans, to a

single comparison at the equal weight: accept iff $(s-1)^\theta + (s+1)^\theta \geq 2$ for $\theta > 0$, iff $s^2 \geq 2$ for $\theta = 0$, and iff $(s-1)^\theta + (s+1)^\theta \leq 2$ for $\theta < 0$. Each rung therefore has an exact critical floor $s^*(\theta)$, and the critical floors form a ladder (Table 7, Figure 4).

Theorem 4 (the substitutability ladder)**.** *On the witness datum:*

(i) (Nesting.) *For rungs $\theta > \theta'$ and every state, acceptance at θ' implies acceptance at θ. The accepted rungs at each state form a prefix of the ladder, so the* critical elasticity $\sigma^*(z) = \inf\{\sigma : z \text{ accepted at } \sigma\}$ *is well defined; it is strictly positive and finite on every gap state with both floors above* 1*, and equal to ∞ on gap states with* $\min(s_1, s_2) \leq 1$*, which only the linear member certifies.*

(ii) (Pointwise.) *For every gap state there is a strictly positive elasticity at which the aggregator rejects it: full Leontief is never necessary at a fixed state.*

(iii) (Uniform.) *For every $\theta < 1$ the rung falsely certifies an exact interval $[s^*(\theta), 2)$ of diagonal gap states, with $s^*(\theta) < 2$ strictly; the critical floors are strictly increasing in depth, $1 < 5/4 < \sqrt{2} < \varphi < \sqrt{3} < \cdots \to 2$, and converge to the typed boundary. Hence no positive elasticity is uniformly safe on the gap region,* $\inf_z \sigma^*(z) = 0$*, and the only uniformly safe member is $\sigma = 0$ – the Leontief member, which is exactly the typed operator ($\mathcal{V}^0 = \mathcal{V}_{\text{typ}}$).*

Table 7: The exact ladder on diagonal gap states $z = (x < 1, s, s)$: acceptance condition and critical floor $s^*(\theta)$ per rung.

θ	σ	accept iff	critical floor $s^*(\theta)$
1	∞	$s \geq 1$	1
2/3	3	$s^2 \geq 3$, else $27(s^2-1)^2 \geq (3-s^2)^3$	root in (117/100, 59/50]
1/2	2	$s \geq 5/4$	5/4
0	1	$s^2 \geq 2$	$\sqrt{2}$
-1	1/2	$s^2 \geq s+1$	$\varphi = (1+\sqrt{5})/2$ (8/5 rejects, 13/8 accepts)
-2	1/3	$s^2 \geq 3$	$\sqrt{3}$
$-m$	$1/(m+1)$	$(s-1)^m + (s+1)^m \leq 2(s^2-1)^m$	increasing in m
$-\infty$	0	$s \geq 2$	2: the typed boundary

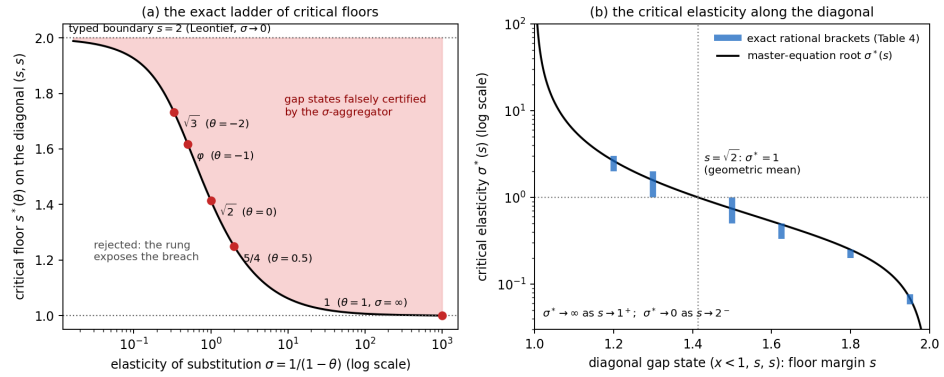


Figure 4: The substitutability ladder (Theorem 4). (a) The critical floor $s^*(\theta)$ on the diagonal as a function of the elasticity σ; the shaded region above the curve is the set of diagonal gap states that the σ-aggregator falsely certifies. The exact algebraic rungs 1, 5/4, $\sqrt{2}$, φ, $\sqrt{3}$ are marked; the curve approaches the typed boundary $s = 2$ only as $\sigma \to 0$. (b) The critical elasticity $\sigma^*(s)$ along the diagonal (root of the master comparison), with the exact rational brackets of the deposited verifier overlaid: at the witness point $s = 6/5$, $\sigma^* \in (2, 3)$; at $s = 5/4$, $\sigma^* = 2$ exactly; $\sigma^* \to \infty$ as $s \to 1^+$ and $\sigma^* \to 0$ as $s \to 2^-$.

The theorem settles the objection in both directions. Pointwise, the separation is not a Leontief phenomenon: at the witness point every aggregator with $\sigma \leq 2$ already rejects, and the dashboard that reports $(1/5 + 11/5)/2 = 6/5 \geq 1$ at the trough is contradicted by the geometric index $\sqrt{11/25} < 1$ on the same datum. Uniformly, however, limited substitutability does not restore soundness: for every positive elasticity there are gap states, arbitrarily close to the typed boundary, that the aggregator certifies and no plan completes safely. The appearance of $\sqrt{2}$, φ and $\sqrt{3}$ as the exact critical floors of the geometric, harmonic and $\sigma = 1/3$ members is, to our knowledge, a new exact witness for the substitutability debate. Off the diagonal the geometric member's acceptance region is $\{s_2 \geq \rho(s_1)\}$ for an explicit decreasing involution ρ with fixed point $\sqrt{2}$, and the complete two-dimensional landscape of σ^* – including the fact that, at a fixed total margin below $2\sqrt{2}$, a moderately concentrated deficit is *more* certifiable than a balanced one – is decided by certified rational enclosures in SI S3. The identification of the $\theta = 0$ member with the Living Planet Index is at the level of functional form only; the index's published inputs and weighting are not Protocol 2, and the rung decision is the adversarial version of its reading (SI S3.5).

4.6. Beyond action-indexed disturbances: the common shock

A second objection is that the gap depends on the action-indexed disturbance convention, under which the adverse event strikes only the coordinate the active plan moves. Decompose the datum as on the benchmark: each principal plan's announced schedule is a mid-interval tent of depth $3/2$ in its characteristic coordinate, and each environmental event is a transient tent dip of depth δ in one floor coordinate (the datum's depth $2 = 3/2 + 1/2$ fixes $\delta = 1/2$); STAGED's floors grow linearly, so an event of depth δ dips a growing floor by $\max(0, \delta - 1/8)$. Under the *common-shock class* the two coordinate events are separate disturbances each of which strikes every plan – a marine heatwave striking the same biomass coordinate regardless of which harvest plan is running.

Proposition 6 (the common-shock verdict)**.** *Under the common-shock class at $\delta = 1/2$:*

(i) $\mathcal{V}^{\mathrm{cs}}_{\mathrm{typ}} = \{s_1 \geq 2,\ s_2 \geq \frac{1}{2}\} \cup \{s_2 \geq 2,\ s_1 \geq \frac{1}{2}\} \cup \{x \geq 1,\ s_1 \geq \frac{3}{8},\ s_2 \geq \frac{3}{8}\}$ *and* $\mathcal{V}^{\mathrm{cs}}_{\mathrm{weak}} = \{x < 1,\ s_1 \geq \frac{1}{2},\ s_2 \geq \frac{1}{2},\ s_1 + s_2 \geq 2\} \cup \{x \geq 1,\ s_1 \geq \frac{3}{8},\ s_2 \geq \frac{3}{8}\}$*, so the gap* $\mathrm{FP}_{\mathrm{cs}} = \{x < 1,\ \frac{1}{2} \leq s_1 < 2,\ \frac{1}{2} \leq s_2 < 2,\ s_1 + s_2 \geq 2\}$ *survives: it is the action-indexed gap I reduced by the exact $1/2$-cut, nonempty with interior, and the witness point is a member. On the surviving region the licensing thresholds are unshifted.*

(ii) *The rescue threshold becomes a trichotomy: $\kappa^*_{\mathrm{cs}} = 0$ on $\mathcal{V}^{\mathrm{cs}}_{\mathrm{typ}}$; $(1 - x)_+$ on failure states with $\min(s_1, s_2) \geq \frac{3}{8}$; and $+\infty$ on failure states with $\min(s_1, s_2) < \frac{3}{8}$, which no reserve increment rescues (witness $(\frac{1}{2}, \frac{1}{4}, 3)$).*

(iii) *The blend window is nonempty exactly on $\{s_1 \geq \frac{1}{2},\ s_2 \geq \frac{1}{2},\ s_1 + s_2 \geq \frac{5}{2}\}$, so the common-shock gap is not blend-closed on the band $s_1 + s_2 \in [2, \frac{5}{2})$ (witness $(\frac{1}{2}, 1, 1)$).*

The separation is therefore not an artefact of the action-indexed convention; what the shared shock changes is the remedy. On the $1/2$-cut the compensatory verdict itself flips from accept to reject, and on $\{\min(s) < 3/8\}$ the prescription changes from accumulating reserve to rebuilding the weaker floor first, because no reserve increment certifies. The coupled class (one disturbance carrying both events), its depth dependence with exact boundary $\delta = 5/4$, the relocation of the gap to larger floors at full magnitude, and the collapse of the separation under a replace reading in which every plan faces the same plan- and weight-independent dip are stated and proved in SI S4; the mechanism of that collapse is instructive – when every plan faces the

same dip, weight-wise plan choice buys nothing, and the per-weight cover degenerates to the typed condition.

4.7. Verification

Every continuum statement above is proved: Lemmas 1–2, Propositions 1–2 and Theorem 1 in Appendix A; Theorem 2, Theorem 3 and Propositions 4–5 in Appendix B; Proposition 3, Theorem 4 and Proposition 6 in SI S2. Three further layers check the finite rational content. (a) A deterministic exact-integer grid verifier (scale 40; no floating point, tolerances or randomness) checks the plan classifications, the region identities and the accepted-set identities of Theorem 2 and Proposition 3(i) on a $31^3 = 29{,}791$-state grid over (x, s_1, s_2), with the critical ratios ρ_1, ρ_2 and their midpoint tested at every grid state; its 25 checks are enumerated in SI S7. (b) Separate fail-loud exact-rational verifiers (standard-library fractions only, byte-reproducible logs) check the ladder and its off-diagonal closure (57, 50 and 44 checks), the common-shock variant (77 checks, with a handshake that reproduces Theorem 2 under the action-indexed class on 676 grid states) and the benchmark of Section 5 (24 checks); the lists are not pooled. (c) The theorem layer is formalised in Lean 4: the witness datum exactly, Theorem 2(1)–(7), Lemma 2 and Proposition 1 on the datum, Theorem 3 and Proposition 4, the handshake and collapse conventions, the master-equation reductions and Theorem 4(ii)–(iii) with its false-certification witnesses, in a module of 199 theorems that compiles with no `sorry`, no additional axioms and no computer-algebra oracle, at an ordered-field interface of which the real-number statements are an instance. The Lean layer and the appendix proofs were produced independently and agree; neither replaces the other, and the grid check is a third and separate thing again, since it validates the symbolic classification on one enumerated instance, which is all a finite check can do. Code, logs and figure pipelines are deposited (Abaee, 2026d).

5. A resource-transition reading of the datum

The witness datum is abstract by design. To show that the theorems bear on a textbook dynamical system, and to let each symbol be read as a quantity an agency monitors, we realise the datum as a Schaefer (1954) biomass–yield model. The realisation is an exact rational re-reading, not a fitted case study: every number is re-derived by the deposited benchmark script, and no empirical claim is made (full detail in SI S5).

The system. Let $B(t)$ be spawning biomass (kt) over one annual review period, with logistic surplus production $\varsigma(B) = rB(1 - B/K)$, $r = 4$, $K = 10$, and a biomass limit $B_{\lim} = 2$ kt below which recruitment is judged impaired. The ecological floor is the margin $s_1 = B - B_{\lim} \geq 0$; the income floor $s_2 \geq 0$ is the fleet income margin net of costs, maintained by the quota and by a transition reserve x (an adjustment fund financing fleet drawdown). The four plans are standard in transition management: NO-SWITCH (status-quo harvest), FAST (an immediate large quota cut – a pulse season followed by a closure – accepting a mid-season trough on the ecological floor), SLOW (a gradual reduction shifting the burden to the income floor) and STAGED (a below-sustained-yield quota while the reserve finances the fleet transition, after which both margins improve). The disturbance is a mid-season heatwave strike imposing an excess mortality $\delta_0 = 1/2$ kt, present or absent. The state is observed through the institutional dashboard $w \cdot (s_1, s_2)$, and the reviewer checks the dashboard rather than the floors – a deliberately adverse reporting architecture adopted to isolate the operator issue, not a description of current fisheries practice, in which a limit-reference-point breach triggers a mandatory response.

The schedule, exactly. At the witness point the stock is $B = 16/5$ kt. FAST is realised by the quota $H^*(t) = 3 + \varsigma(B_{\text{line}}(t))$ on $(0, \frac{1}{2})$ – a pulse declining from $1463/125 \approx 11.7$ to $2161/250 \approx 8.6$ kt/yr, admissible against a fleet capacity of 13 – followed by a closure, $H^* = 0$. The benign branch tracks the tube line $s_1 : \ 6/5 \to -3/10 \to 6/5$ exactly; the heatwave drops the trough from $17/10$ to $6/5$ kt, the adverse tube's value, and on the recovery leg the surplus is at least $\varsigma(6/5) = 528/125 > 4$ kt/yr while the certified recovery needs only 4, so the realised nonlinear trajectory lies on or above the certified tube throughout: the certified piecewise-linear tubes are conservative for the Schaefer realisation, by an exact inequality that uses the monotonicity of ς below $K/2$. NO-SWITCH and SLOW hold $B = 16/5$ at the sustained yield $\varsigma(16/5) = 1088/125 \approx 8.7$ kt/yr (below the model's maximum sustainable yield $rK/4 = 10$); STAGED, entered at the rescue witness with fund $x = 3/2$, rebuilds the stock $16/5 \to 69/20$ while the fund draws down $3/2 \to 1/2$ and both margins improve by $e = 1/4$.

The dashboard reading. Under the heatwave branch of the licensed plan the equal-weight index $s_1 + s_2$ takes the tube values $(12/5,\ 2/5,\ 12/5)$: the dashboard never drops below $2/5$, and the transition is certified at every weighting (the thresholds $\rho_1 = 2/3$ and $\rho_2 = 3/2$ place FAST in every income-heavy assessor's acceptance set and SLOW in every biomass-heavy assessor's).

The biomass floor, meanwhile, is breached mid-transition on both branches ($-4/5$ with the strike, $-3/10$ without): the dashboard certifies a transition that violates the biomass limit whether or not the heatwave occurs, and no weighting can detect this from the index alone (Figure 5). The operational number is the reserve threshold: a fund below the buy-back cost $c = 1$ makes the staged transition unfinanceable, so every state of the failure set carries the shortfall $\kappa^* = 1 - x$, and a top-up of the adjustment fund converts an index-licensed impossibility into a rescue. In management language, the dashboard is structurally unable to distinguish a transition that survives the heatwave from one that does not; the financing gap, not the weighting, is what separates them.

Decision-analytic structure. Read as a decision problem the benchmark is a sorting problem in multi-criteria decision aiding: the decision maker is the resource authority; the alternatives are the four management schedules; the criteria are the two floors, evaluated along the whole transition under the declared disturbance; and the weights are the preference orderings of the doctrine under which the index is read. The criteria conflict – each principal schedule protects one at the expense of the other – and each is defended by a distinct constituency, so the weight space reads both as an assessor's calibration and as the stakeholder dispute. The deliverables are ones a decision analyst can communicate: the licensing thresholds (the preference range over which each alternative is certified), the rescue threshold (the adjustment funding that converts an unlicensable state into a staged transition), and, from Theorem 4, the elasticity below which an aggregator exposes the breach ($\sigma^* \in (2, 3)$ at the witness point: every $\sigma \geq 3$ aggregator certifies with no mid-transition response, every $\sigma \leq 2$ member rejects and triggers the closure-and-rebuilding response). The aggregator choice – a reporting decision – flips a management action.

Units on a real stock. The units are anchored to a regulated floor: for Northern cod (NAFO Divisions 2J3KL) the assessed spawning-stock series carries a limit reference point of 884.6 kt (DFO, 2016), and reading the benchmark's floor as this LRP places the witness's opening stock (1.6 LRP units, ≈ 1.42 Mt) in the neighbourhood of the assessed mid-1980s biomass and the adverse trough (0.6 LRP units, ≈ 531 kt) below the LRP but above the 2015 assessed level of about a third of it; the adverse plan's worst-case excursion, one full LRP unit, is of the order of the stock's historical swing across the LRP. The anchoring fixes units only: no parameter of the datum is estimated from data, and the companion scored forecast test on the

same series (Abaee, 2026c) is a caution of exactly this paper's discipline – no structural surplus-production model in its ladder beat last-value persistence, so what survives scrutiny on such a stock is the menu and the floors, both set by management and both taken as given here.

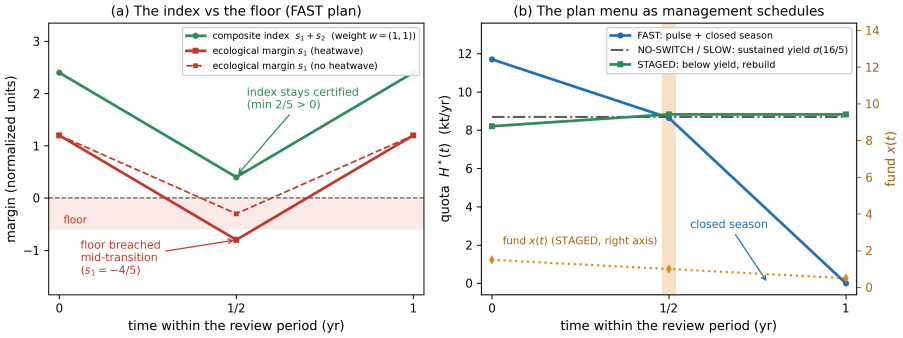


Figure 5: The witness datum realised as a fishery transition. (a) Under the pulse-and-closure plan the composite index $s_1 + s_2$ at $w = (1, 1)$ remains certified at every instant (minimum $2/5$) while the biomass margin breaches its floor mid-transition ($-4/5$ with the heatwave strike, $-3/10$ without). (b) The plan menu as management schedules: pulse plus closed season; sustained-yield status quo; and the reserve-financed staged transition, which rebuilds the stock while improving both margins. All values are the exact rationals of SI S5.

6. Discussion

6.1. What the separation is, and what it is not

The scalarized operators $\{E_w\}_{w \in W_+}$ formalise *one* compensatory doctrine: a single aggregate index, nonnegative weights on the floors, substitution across floors permitted at those weights, disturbances respected. The weak-sustainability literature is broader than this operator – it encompasses aggregate wealth, nondeclining comprehensive consumption, shadow prices and intertemporal investment conditions (Solow, 1974; Asheim, 1994; Dasgupta and Mäler, 2000; Neumayer, 2013) – and strong sustainability is not always the requirement that every stock coordinate stay nonnegative at every instant; critical-natural-capital and safe-operating-space frameworks use thresholds, irreversibility and resilience (Ekins et al., 2003; Rockström et al., 2009; Raworth, 2012; Doyen and Gajardo, 2020). The typed operator idealises the separately-binding-floor reading in the same sense in which the scalarized operators idealise one aggregation doctrine. Within this scope, Theorem 2 says that the two formalised doctrines can disagree on the same

datum, with the same robustness standard and the same menu, in the direction compensatory-accepts/noncompensatory-rejects, on a relatively open region; that the disagreement is not an artefact of one bad weight, since every weight accepts, each licensing a different physical transition; and that its exact seat, by Proposition 1(ii), is the noncommutation of "choose a plan" with "for all weights". The two quantifier orders carry an information reading: $\exists a\, \forall w$ is a commitment made before the weight is known, $\forall w\, \exists a_w$ lets the plan be chosen after it is observed, and on the witness the gap measures the value of information about the assessment weight. Theorem 3 qualifies the reading: since one convexified blend serves every weight on I, that value reflects the finite deterministic menu rather than the weight information itself, and Proposition 4 adds that it is removed by convexifying the menu, not by sharing plans in time.

Several things are not claimed. The theorem establishes no ranking of doctrines, and Theorem 3 shows that the structural character of the separation is a property of the menu, not of either tradition. There is no separation on every datum: where one plan is safe for all weights the assessments coincide. The weights model an assessment convention, not prices, and nothing is claimed about welfare. $\mathcal{V}_{\text{weak}}$ is the strictest compensatory reading (Proposition 2), not the genuine-savings or inclusive-wealth criterion; the witness's reserve is excluded from the aggregate by construction, isolating the case in which the index measures the service floors while the reserve is a separate instrument – had the reserve entered the aggregate with positive weight the gap could shrink or disappear. The impossibility region is a negative certificate relative to the four-plan menu; with a larger menu it can shrink or vanish, and menu completeness is a declaration the application must make. The separation is exhibited for the action-indexed disturbance class and the common-shock class (Proposition 6); under coupled all-floor shocks its status is reading-conditional and depth-dependent (SI S4), and under a replace reading in which every plan faces the same plan- and weight-independent dip the typed/weak distinction itself collapses – a regime that evidence on interacting planetary-boundary processes (Lade et al., 2020) suggests is not remote. Finally, all statements are for finite-horizon exact-tube data: infinite horizons, stochastic constraints, partial observation and endogenous event times are outside the scope.

6.2. Acceptance semantics in multi-criteria decision aiding

For decision aiding the protocols of Table 1 are sorting rules, and the separation is a statement about which rule a composite index can support.

Three readings connect it to established practice. First, the licensing thresholds of Theorem 2(6) are the verdict's exact *weight-sensitivity intervals* – the preference ranges over which each alternative is certified – the acceptance-side analogue of the sensitivity analyses applied to composite-indicator rankings (Nardo et al., 2008; Becker et al., 2017), with the difference that here the sensitivity is to *which plan* is licensed, not merely to a rank. Second, the compensability analyses of multi-criteria methods for sustainability assessment (Cinelli et al., 2014) and of outranking methods (Schär et al., 2025) classify aggregation behaviour at a fixed profile; the separation acts one level up, in the acceptance protocol over a transition, and arises even where every static level of the index is reported correctly (Lemma 2). Third, the here-and-now/wait-and-see structure of adjustable robust optimisation (Ben-Tal et al., 2004) gives the gap its decision-theoretic name: common-plan acceptance is the static robust problem, weight-adaptive acceptance the adjustable one, and Theorem 1 says that on a finite menu the adjustable problem is the convex relaxation of the static one – the pure/mixed duality gap, in the game form. The maximin and viability formulations of strong sustainability (Martinet, 2011; Cairns and Martinet, 2014; Doyen and Gajardo, 2020) and the weak-comparability foundation of ecological economics (Martinez-Alier et al., 1998) are the positions the typed operator formalises; the per-literature notes are in SI S8. Two companion studies bracket the present result within the same programme: a two-land biocapacity model in which a weighted aggregate meets its floor while the typed floor on ecological capital is violated – a continuous-time instance of the compensatory gap, for which Theorem 4 is the aggregator-side statement (Abaee, 2026b) – and the obstruction calculus whose substitution-pathway certificates are separation certificates rather than universal exchange rates, of which the ladder is the elasticity-side statement (Abaee, 2026a).

6.3. Design tests for composite indicators

A composite indicator can be statically lossless yet dynamically unsound as a transition-certification device. For indicator developers the results convert into four design tests, each a reading of a proved statement rather than a heuristic.

1. **Static losslessness.** Does the aggregate preserve component violations at a fixed trajectory for the declared weight family? At the full cone, yes (Lemma 2); for a restricted family the test is to characterise the violations the family can miss (Proposition 2).

2. **Protocol quantifiers.** Does the certification rule require a common plan or permit weight-adaptive plan selection? If the latter, the indicator is exposed to the gap on exactly the region Theorems 1–2 exhibit, and the plan licensed in each weight range should be reported with the verdict.

3. **Path-wise observability.** Is the index evaluated along the full tube or only at audited endpoints? Endpoint evaluation is the weakest operator of the chain and cannot detect transient breaches: on the witness, FAST is typed-endpoint-admissible at every state of X_0 (its endpoint s_1 equals its initial s_1) but typed-tube-admissible only for $s_1 \geq 2$.

4. **Menu completeness and convexity.** Is the plan menu exhaustive, and are blended plans implementable? Negative verdicts are menu-relative, and the gap is a property of the deterministic menu that convexification collapses exactly (Theorem 3); the availability of the convexified menu is external data the documentation must declare.

Where separately binding floors matter, the conclusion is a detection requirement rather than a presentation preference: component trajectories and their minimum margins must be reported alongside the aggregate, so that the typed verdict can be reconstructed and the rescue set told from the impossibility region. The same separately-checked-floor logic underlies the safe-and-just-space literature, which evaluates social thresholds and biophysical ceilings without cross-compensation (Raworth, 2012; O'Neill et al., 2018; Fanning et al., 2022). Where a resource-controlled rescue plan exists, the minimum increment κ^* is the number to report, and investment in that margin – not reweighting – is the candidate remedy. A ten-point audit checklist, a minimum report, the data requirements for empirical application and a terminology map for readers from operations research, control theory and environmental governance are given in SI S6.

6.4. Limitations

(i) The separation is an existence result with an open region; it does not imply that the gap is large in any application, and where the assessments coincide it yields nothing.

(ii) The finite-menu theorem governs plans in the margin class; reserve-financed and destination-failing plans are handled separately, and the geometric form of the gap is established within that class.

(iii) If a convexified plan is physically and institutionally available the gap disappears (Theorem 3); the separation is a statement about certification relative to the available menu.

(iv) The operators cover finite horizons with exact tubes and specified disturbance sets; the persistence result is hold-prefix specific, and arbitrary multi-stage extension can erase the gap (Proposition 3(ii)).

(v) The conservatism of the benchmark's certified tubes is regime-conditional: it uses the monotonicity of the Schaefer surplus below $K/2$, and depensation or nonlinearly amplified heatwave mortality at low stock would reduce the recovery surplus; the certification statements, which concern the datum rather than the realisation, are unaffected.

(vi) In the ladder, interior critical elasticities and the off-diagonal boundary curve are transcendental and carry certified rational brackets, not closed forms; the common-shock statements are proved on the datum's decomposition and verified across a δ-family, and no claim is made beyond it.

(vii) Computationally, the backward recursion is polynomial on a finite explicit state–action–disturbance graph, but continuous, hybrid or belief-state models may make exact-tube inclusion hard or undecidable, and grid cardinality grows exponentially in the number of floors; the witness is tractable because its datum is small, finite and rational.

(viii) Novelty statements reflect a bounded literature search; no empirical claim is made; and the framework extensions deposited with the paper (SI S9) are at partial status and are not used in any proof.

7. Conclusions

Scalarized and coordinate-wise feasibility criteria are usually compared as doctrines – as normative stances on substitutability, of which the weak- and strong-sustainability traditions are the canonical instances. This paper shows that their divergence survives translation into assessment mechanics, as a theorem about two operators on a common datum. At any fixed trajectory the full-cone aggregate is lossless, so the divergence is not scalarization blindness; it is the failure of "some plan for every weighting" to commute with "one plan for all weightings". For any finite menu of margin-class plans the acceptance gap is the part of the convex hull of the required margins

that no single plan dominates, so it is a property of the finiteness and non-convexity of the menu rather than of either doctrine; on an explicit rational datum it is a region with nonempty interior whose states split, at a closed-form reserve threshold, into those a staged plan can rescue and those no available plan completes safely. Blending the plans closes the gap exactly, alternating them in time does not, and neither limited substitutability nor a shared shock removes it: every positive-elasticity aggregator falsely certifies an exact interval of gap states, with the typed boundary approached only in the Leontief limit, and under a common shock the gap survives reduced by an exact cut while the rescue route fails on an exact band.

For the construction and use of composite indices the consequence is concrete. An index can be sound as accounting and still over-certify as assessment, because certification is a quantifier statement about transitions, not a property of a number. Where separately binding floors matter, the aggregate alone cannot distinguish the rescue region from the impossibility region, which differ only in a resource the index does not see; what a composite index needs in order to certify a transition is a single plan that serves every admissible weighting – membership in $\mathcal{V}_{\mathrm{typ}}$ – and that plan is in general a convexified plan rather than a member of the deterministic menu, so the question a reporting convention must answer is whether its action set admits the convexification. Per-floor reporting along the path is the detection requirement this makes exact; the licensing thresholds, the rescue threshold and the critical elasticity are the numbers a decision analyst can hand to an authority with the verdict.

Appendix A. Proofs of the general results

Proof of Lemma 1. If $a \in \bigcap_\lambda E_\lambda(z)$ then each $E_\lambda(z)$ is nonempty, so z lies in the right-hand set. The inclusion fails at z precisely when each $E_\lambda(z)$ is nonempty but $\bigcap_\lambda E_\lambda(z) = \varnothing$, i.e. when no common selector exists, which is the stated equality condition. □

Proof of Lemma 2. ($\Rightarrow$) $w \geq 0$, $w \neq 0$, $v \geq 0$ give $w \cdot v = \sum_i w_i v_i \geq 0$. ($\Leftarrow$) If $v_k < 0$, take $w = e_k \in W_+$ (the k-th standard basis vector): $w \cdot v = v_k < 0$. □

Proof of Proposition 1. (i) Let $a \in E_{\mathrm{typ}}(z)$. For every d, $\mathrm{Tube}(a, d) \subseteq S = S^{\mathrm{phys}} \cap \{s \geq 0\}$; by Lemma 2 every tube point satisfies $w \cdot s \geq 0$ for every $w \in W_+$, and likewise every successor lies in G^w, so $a \in E_w(z)$. The remaining inclusions follow from $S^w \subseteq S^{\mathrm{phys}}$, $G^w \subseteq G^{\mathrm{phys}}$ and $\mathrm{End}(a, d) \subseteq$

Tube(a,d). (ii) The inclusion $\subseteq$ is (i) intersected over w. Conversely let $a \in \bigcap_{w\in W_+} E_w(z)$. For every d and every tube point p, $w\cdot s(p) \geq 0$ for every $w \in W_+$, so $s(p) \geq 0$ componentwise by Lemma 2; hence Tube$(a,d) \subseteq S$, and the same argument with Succ gives Succ$(a,d) \subseteq G$. Thus $a \in E_{\mathrm{typ}}(z)$. □

Proof of Proposition 2. By Proposition 1(i), $E_w(z) \subseteq E_{\mathrm{tube,phys}}(z)$ for every w, so $\mathcal{V}_w \subseteq \mathcal{V}_{\mathrm{phys}}$ and the intersection over W preserves the inclusion. If $z \in \mathcal{V}_{\mathrm{typ}}$, Proposition 1(ii) gives $a \in \bigcap_{w\in W_+} E_w(z) \subseteq \bigcap_{w\in W} E_w(z)$, so $E_w(z) \neq \varnothing$ for every $w \in W$, i.e. $z \in \mathcal{V}_W$. Monotonicity: intersecting over a larger family can only shrink the intersection. □

Proof of Theorem 1. (i) is the margin test restated. For (ii), if $s = d + v$ with $d \in \mathrm{conv}(\mathcal{D})$ and $v \geq 0$, then for every $w \geq 0$, $w\cdot s \geq w\cdot d \geq \min_a w\cdot d_a$, so the scalarized test holds. Conversely suppose $s \notin \mathrm{conv}(\mathcal{D}) + \mathbb{R}^n_+$. This set is closed and convex (a compact polytope plus a closed cone), so strict separation gives $w \neq 0$ with $w\cdot s < w\cdot d + w\cdot v$ for every $d \in \mathrm{conv}(\mathcal{D})$ and every $v \geq 0$. Since v ranges over the whole cone, $w \in \mathbb{R}^n_+ \setminus \{0\}$; taking $v = 0$ gives $w\cdot s < \min_{d\in\mathrm{conv}(\mathcal{D})} w\cdot d = \min_a w\cdot d_a$, so the scalarized test fails at w. The gap formula is (ii) minus (i); if $\mathcal{D}$ is replaced by $\mathrm{conv}(\mathcal{D})$ the union in (i) becomes $\mathrm{conv}(\mathcal{D}) + \mathbb{R}^n_+$ and the gap is empty. □

Appendix B. Proofs for the witness datum

Proof of Theorem 2. **(1)** Typed admissibility requires the worst-case tube in S_0 and the successor in G. NO-SWITCH never reaches G (its successor retains $q = 0$). For FAST the worst-case s_1-tube is $[s_1 - 2, s_1]$, safe iff $s_1 \geq 2$; its successor $(1, x, s + e) \in G$ given $x \geq 0$. Symmetrically SLOW is typed-admissible iff $s_2 \geq 2$. For STAGED the floors grow and the successor lies in G; the binding constraint is the x-tube $[x - 1, x]$, safe iff $x \geq 1$. Hence $\mathcal{V}_{\mathrm{typ}} = \{x \geq 1\} \cup \{s_1 \geq 2\} \cup \{s_2 \geq 2\}$.

(2) Fix $w = (w_1, w_2) \in W_+$. NO-SWITCH misses G^w. FAST is w-admissible iff its worst-case aggregate stays nonnegative, $w_1(s_1-2)+w_2s_2 \geq 0$, i.e. $w_1s_1+w_2s_2 \geq 2w_1$; symmetrically SLOW iff $w_1s_1+w_2s_2 \geq 2w_2$; STAGED iff $x \geq 1$, since $w\cdot s \geq 0$ holds automatically along its tube. Hence $z \in \mathcal{V}_w$ whenever $x \geq 1$; if $x < 1$, $z \in \mathcal{V}_w$ iff one of the two inequalities holds. If $x \geq 1$ then $z \in \mathcal{V}_{\mathrm{weak}}$. If $x < 1$ and, say, $s_1 \geq 2$, FAST is typed-admissible by (1), hence w-admissible for every w, so $z \in \mathcal{V}_{\mathrm{weak}}$. It remains to treat $x < 1$, $s_1 < 2$, $s_2 < 2$. For $w_1 > 0$ write $r = w_2/w_1$: FAST is licensed iff $s_1 + rs_2 \geq 2$, i.e. $r \geq (2-s_1)/s_2 = \rho_1$ (for $s_2 > 0$; when $s_2 = 0$ the condition is $s_1 \geq 2$, excluded), and SLOW iff $s_1 + rs_2 \geq 2r$, i.e. $r \leq s_1/(2 - s_2) = \rho_2$.

At $w_1 = 0$ ($r \to \infty$) FAST is licensed provided $s_2 \geq 0$ while SLOW is rejected on $s_2 < 2$; at $w_2 = 0$ ($r = 0$) symmetrically. Therefore $z \in \mathcal{V}_w$ for every w iff $\{r \geq \rho_1\} \cup \{r \leq \rho_2\} = [0, \infty]$, i.e. iff $\rho_2 \geq \rho_1$, and

$$\frac{s_1}{2 - s_2} \geq \frac{2 - s_1}{s_2} \iff s_1 s_2 \geq (2 - s_1)(2 - s_2) \iff s_1 + s_2 \geq 2,$$

both denominators being positive. Assembling the cases, and noting $\{s_1 \geq 2\} \cup \{s_2 \geq 2\} \subseteq \{s_1 + s_2 \geq 2\}$ under nonnegativity, $\mathcal{V}_{\text{weak}} = \{x \geq 1\} \cup \{s_1 + s_2 \geq 2\}$.

(3) The physical operators constrain only $x \geq 0$ (with $q = 1$ at the destination). FAST and SLOW have tubes in S^{phys} and successors in G^{phys} for every $z \in X_0$ (their dips affect only s), so $\mathcal{V}_{\text{phys}} = X_0$; their endpoint values likewise, so $\mathcal{V}_{\text{end}} = X_0$.

(4) By (1)–(2), $\text{FP}_{\text{agg}} = [\{x \geq 1\} \cup \{s_1 + s_2 \geq 2\}] \setminus [\{x \geq 1\} \cup \{s_1 \geq 2\} \cup \{s_2 \geq 2\}] = \{x < 1,\ s_1 < 2,\ s_2 < 2,\ s_1 + s_2 \geq 2\} = I$, nonempty (e.g. $(\frac{1}{2}, \frac{6}{5}, \frac{6}{5})$) with the stated interior; $R \subseteq \{x \geq 1\} \subseteq \mathcal{V}_{\text{typ}}$.

(5) The first strictness is (4). The point $(\frac{1}{2}, \frac{1}{10}, \frac{1}{10})$ has $x < 1$ and $s_1 + s_2 = \frac{1}{5} < 2$, so it is outside $\mathcal{V}_{\text{weak}}$ by (2) and inside $\mathcal{V}_{\text{phys}}$ by (3).

(6) The threshold characterisation was derived in (2). On $\mathcal{Q}$, $s_1 + s_2 \geq 2$, so the two licensing intervals overlap and cover all weights; on the relative interior ($s_1 + s_2 > 2$) both thresholds are finite and $r > \rho_2$ licenses FAST only, $r < \rho_1$ SLOW only. On I ($x < 1$) STAGED is inadmissible for every w, so $E_{\text{typ}}(z) = \bigcap_w E_w(z) = \varnothing$ by Proposition 1(ii). On R ($x \geq 1$) STAGED lies in $E_{\text{typ}}(z)$ and serves every weight.

(7) On R, STAGED's tube is $[x - 1, x] \times \{s \to s + e\}$ with $x \geq 1$, the floors grow, and the successor $(1, x - 1, s + e) \in G$. On I the four violations exhaust the menu: FAST's s_1-tube reaches $s_1 - 2 < 0$; SLOW's s_2-tube reaches $s_2 - 2 < 0$; STAGED's x-tube reaches $x - 1 < 0$; NO-SWITCH's successor misses G. □

Proof of Theorem 3. (i) FAST's worst-case tube has s_1-coordinates $[s_1 - 2, s_1]$ and s_2-coordinate $\{s_2\}$; SLOW's is the mirror; both keep x nonnegative. The blended tube therefore has s_1-coordinates $[s_1 - 2\delta, s_1]$, s_2-coordinates $[s_2 - 2(1 - \delta), s_2]$, and nonnegative x. Typed admissibility requires the tube within S_0 and the successor in G; the tube condition is exactly $\delta \leq s_1/2$ and $\delta \geq 1 - s_2/2$, and the blended successor is the mixture of two equal successors $(1, x, s + e) \in G$. The window is nonempty iff $1 - s_2/2 \leq s_1/2$, i.e. $s_1 + s_2 \geq 2$; on $\mathcal{Q}$ it lies in $(0, 1)$ automatically, and at equality it is the singleton $\delta = s_1/2$, feasible because S_0 is closed. (ii) If $s_1 + s_2 \geq 2$, (i) supplies a typed-admissible blend for every $x \geq 0$; if $s_1 + s_2 < 2$ no blend

is typed-admissible, and the deterministic menu contributes $\mathcal{V}_{\text{typ}} \subseteq \{x \geq 1\} \cup \{s_1 + s_2 \geq 2\}$. Hence the augmented typed set is $\{x \geq 1\} \cup \{s_1 + s_2 \geq 2\} = \mathcal{V}_{\text{weak}}$ by Theorem 2(2), and nothing beyond. (iii) Immediate from the window's dependence on (s_1, s_2) alone and from typed admissibility implying w-admissibility for every weight. □

Proof of Proposition 4. A policy that spends positive time on FAST visits every $s_1 \in [s_1 - 2, s_1]$, and positive time on SLOW every $s_2 \in [s_2 - 2, s_2]$; the union tube lies within S_0 exactly when both dips are nonnegative, i.e. $s_1 \geq 2$ and $s_2 \geq 2$. On I both coordinates are strictly below 2, so no such policy is typed-admissible. □

Proof of Proposition 5. Only STAGED_κ depends on κ, and its typed admissibility is independent of the floors: its x-tube is $[x + \kappa - 1, x + \kappa]$, the floors are nondecreasing, and the successor $(1, x + \kappa - 1, s + e) \in G$ exactly when $x + \kappa \geq 1$. If $z \in \mathcal{V}_{\text{typ}}$ some base plan is admissible at $\kappa = 0$, so $\kappa^*(z) = 0$. If $z \notin \mathcal{V}_{\text{typ}}$ – $x < 1$, $s_1 < 2$, $s_2 < 2$ – then NO-SWITCH, FAST and SLOW all fail by Theorem 2(1), the only admissible element of $\mathcal{A}_\kappa$ is STAGED_κ, admissible exactly for $\kappa \geq 1 - x$; hence $\kappa^*(z) = 1 - x$. □

The proofs of Proposition 3, Theorem 4 and Proposition 6 are given in SI S2; the master-equation reduction behind Theorem 4 is the averaging argument that the two principal plans' acceptance conditions are affine in the weight share and sum to a weight-independent quantity, so that the equal weight is always the binding one on the diagonal.

Declarations

Availability of data and materials. All code and results are available in the figshare deposit (Abaee, 2026d): the exact-integer grid verifier (25 checks), the exact-rational verifiers for the substitutability ladder, its off-diagonal and continuum closures and the common-shock variant, the benchmark script (24 checks), the figure pipelines, and the preserved novelty-search strings; the Lean 4 formalisation is in the programme repository (`github.com/MIKEAA2020/general-sustainability`, directory `lean/`). No empirical data were generated or analysed.

Competing interests. The author declares no competing interests.

Funding. Not applicable.

Author contributions. A.A. conceived the work and wrote, edited and reviewed the manuscript.

AI declaration. GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review; all results, proofs and verifications were checked by the author.

Supplementary Information

The Supplementary Information (SI) contains: S1 abbreviations; S2 remaining proofs (persistence and erasure, the substitutability ladder, the common-shock verdict); S3 the substitutability spectrum in full (critical-elasticity landscape, off-diagonal criterion, fixed-sum structure, continuum closure, the Living Planet Index identification); S4 the common-shock variant in full (coupled and replace readings, depth family); S5 the resource-transition benchmark in full (schedule, dashboard arithmetic, fisheries translation table, data anchor); S6 practitioner material (terminology map, audit checklist, minimum report, data requirements); S7 the verification record; S8 positioning notes per literature; S9 framework extensions at partial status.

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